Golnaz Hosseini

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1. Education

Pooyesh Institute of Higher Education, Electrical Engineering, Qom, Iran — September 2018 – February 2022

Bachelor in Biomedical Engineering, Bioelectric, GPA: 19.7/20.0 (4.0/4.0)

2. Research Experience

TechnoHealthAI Startup, Tehran, Iran — March 2025 – present

Research project: Towards Efficient Abdominal Multi-Organ Segmentation: A Lightweight Hybrid UNet-SegFormer Network

- Developed HUSF-Net, a lightweight 3D hybrid CNN-transformer framework for abdominal multi-organ segmentation, combining local and global feature representations through multi-stage feature fusion and cross-attention.
- Evaluated the framework across six publicly available abdominal datasets (AMOS-CT, AMOS-MRI, BTCV, FLARE22, WORD, and UW-Madison), covering diverse anatomical structures, imaging modalities, and patient populations.
- Implemented comprehensive preprocessing strategies for each modality to standardize input volumes.
- Examined several representative cases and segmentation failures to identify common challenges and limitations of the model.
- Performed an expert-based qualitative evaluation of 25 unlabeled test cases using radiologist assessment to examine anatomical accuracy and potential clinical applicability.
- Investigated the balance between segmentation accuracy, computational efficiency, robustness across heterogeneous datasets, and reliability of evaluation toward developing a practical framework for clinical deployment.

TechnoHealthAI Startup, Tehran, Iran — January 2025 – November 2025

Research project: Gastrointestinal Organ Segmentation for Radiotherapy Planning

- Reviewed literature on gastrointestinal organ segmentation, medical image analysis, and deep learning for radiotherapy planning.
- Worked with the UW-Madison GI-Tract MRI dataset to segment three organs: small bowel, large bowel, and stomach.
- Preprocessed MRI scans using bias-field correction and intensity normalization; evaluated the effect of bias correction using a K-means-based estimation approach, which showed no measurable improvement in model performance.
- Implemented and compared U-Net, U-Net++, ResUNet, and EfficientUNet++ for gastrointestinal organ segmentation.
- EfficientUNet++ achieved the highest performance, with 87.21% precision, 76.82% recall, 82.45% F1-score, and 69.47% IoU.
- Applied Grad-CAM to visualize model predictions and examine the regions contributing to segmentation outputs.
- Contributed to training interns on medical image segmentation and deep learning workflows as part of a TechnoHealthAI training program.

TechnoHealthAI Startup, Tehran, Iran — April 2024 – September 2024

Research project: Breast Lesion Segmentation

- Reviewed literature on breast MRI, multimodal image analysis, and deep learning-based lesion segmentation
- Worked with the BreaDM dataset, comprising pre-contrast, post-contrast, and subtraction MRI sequences for breast lesion segmentation.
- Preprocessed 3D MRI scans using Gaussian denoising and intensity normalization, and combined the three sequences as input channels for each image slice.
- Implemented a SegResNet model for automated 3D breast lesion segmentation.
- Achieved a Dice score of 79.11%, IoU of 68.54%, and Hausdorff Distance (HD) of 23.04.

Imam Khomeini Hospital, Tehran, Iran — October 2022 – May 2023

Research project: Breast Tumor Segmentation from Multiparametric MRI

- Reviewed literature on breast MRI, multiparametric imaging, and deep learning-based breast tumor segmentation.
- Used a private clinical dataset of multiparametric MRI scans collected at Imam Khomeini Hospital, including pre-contrast and two post-contrast MRI sequences.

- Preprocessed MRI scans using Gaussian filtering, N4 bias-field correction, and intensity normalization. Combined the three MRI sequences into RGB channels, assigning pre-contrast, first post-contrast, and second post-contrast scans to the red, green, and blue channels, respectively.
- Implemented and compared DeepLabV3+, U-Net, PAN, MA-Net, and Mask R-CNN models for breast tumor segmentation.
- Mask R-CNN achieved the highest performance, with 82.66% Dice, 73.86% Jaccard, 80.36% Precision, and 88.22% Recall.

QMISG Research Group, Tehran, Iran — August 2022 – November 2022

Research project: Brain Tumor Segmentation using Multimodal MRI.

- Reviewed literature on deep learning-based brain tumor segmentation and multimodal MRI analysis.
- Preprocessed BraTS 2020 MRI scans using bias-field correction and combined three MRI modalities as three input channels.
- Implemented DeepLabV3+ using the SMP library for segmenting three tumor subregions, including enhancing tumor (ET), tumor core (TC), and whole tumor (WT).

QMISG Research Group, Tehran, Iran — June 2022 – February 2023

Research project: Deep Learning-Based Classification of Benign and Malignant Prostate Lesions from Multiparametric MRI.

- Reviewed relevant literature on prostate MRI, lesion classification, multiparametric MRI, and AI-based medical image analysis.
- Developed a deep learning approach for benign and malignant prostate lesion classification using T2-weighted, ADC, and DWI MRI scans.
- Preprocessed MRI scans from ProstateX and PI-CAI datasets, including registration, resampling, and N4 bias-field correction with SimpleITK and TorchIO.
- Implemented a customized VGG-based classification model with Squeeze-and-Excitation (SE) blocks in PyTorch.

3. Work Experience

Co-founder | TechnoHealthAI Startup, Tehran, Iran — December 2023 – Present

- Conducted AI and deep learning projects focused on medical imaging applications.
- Coordinated with hospitals and clinical specialists on research projects and medical imaging data collection.
- Worked with researchers and clinicians from different fields, including AI, computer engineering, radiology, and oncology.
- Provided research and technical consulting for medical AI projects.
- Trained and mentored interns from Biomedical engineering, computer engineering, and related fields through hands-on medical AI projects.
- Represented TechnoHealthAI at scientific conferences and exhibitions, including ICBME, to engage with researchers and industry professionals.

Research Intern | Quantitative MR Imaging and Spectroscopy Group (QMISG), Tehran, Iran — February 2022 - January 2023

- Collaborated with the research team and hospital staff to collect and organize medical imaging datasets for research.
- Conducted research on the application of deep learning and artificial intelligence to medical image analysis, with a focus on MRI imaging datasets.
- Presented and discussed research progress, methods, and findings within the research group.

Image Processing Intern | National Brain Mapping Laboratory (NBML), Tehran, Iran — July 2020 – August 2020

- Reviewed texts and research literature on MRI physics, MRI image formation, fMRI principles, and neuroimaging methods.
- Studied fMRI experimental design and task paradigms, including the principles of designing task-based fMRI studies.
- Preprocessed a sample finger-tapping fMRI dataset using SPM, gaining practical experience with standard fMRI preprocessing and analysis workflows.
- Reviewed published fMRI studies to understand experimental design, task-based paradigms, and the interpretation of functional neuroimaging data.

Biomedical Engineer Intern| Dadgostary Hospital, Tehran, Iran — April 2020 – June 2020

- Assisted with routine maintenance, troubleshooting, and basic repairs of medical equipment under supervision.
- Provided basic training to nursing staff on the operation and proper use of medical equipment.

4. Volunteer Experience

Conference Volunteer | **ISMRM Iranian Chapter First Annual Meeting**, Tehran, Iran — August 2022

- Supported the registration team with attendee registration, check-in, and participant assistance.
- Assisted with on-site coordination and conference logistics during the event

Member | Aran Student Cultural Club, Pooyesh Institute of Higher Education, Qom, Iran — October 2018 – March 2020

- Collaborated with international students to organize cultural events and commemorative programs.

5. Teaching Experiences

Teaching Assistant – Computer Programming & Systems Analysis | Pooyesh Institute of Higher Education, Qom, Iran — Academic Year 2019 – 2021

- Assisted students with course concepts, assignments, and exam preparation.

6. Skills

6. a. Languages

English: Proficient, TOEFL: Overall 5.5/6.0 (110/120) (Listening 5.5, Reading 5.5, Writing 5.5, Speaking 5.5), 2026; Persian: Native.

6.b. Technical Skills

Programming & ML: Python, PyTorch, MATLAB, scikit-learn, SciPy, OpenCV.

Medical Imaging: Medical Image Analysis, CT/MRI, Image Preprocessing, SimpleITK, MONAI, TorchIO, PyRadiomics.

Research: Deep Learning, Model Development, Explainable AI, Scientific Writing, Literature Review, Research Project Management, Interdisciplinary Collaboration.

7. Certificates

Coursera — September 2022
Deep Learning with PyTorch: Segmentation & Object Detection

DeepLearning.AI (Coursera) — August 2022
AI for Medicine

DeepLearning.AI (Coursera) — July 2022
Advanced Computer Vision with TensorFlow

HowSam — April 2022
PyTorch

FaraDars — January 2022
Python Programming

ACECR, Tehran University of Medical Sciences — April – July 2020
Medical Imaging Equipment

8. Awards and Honors

- **Top Graduate**, BSc Biomedical Engineering, Pooyesh Institute of Higher Education, graduated as the top student in the program, 2022.
- **Top DAFI Scholarship Candidate**, Recognized among top-performing DAFI scholarship recipients, 2022.
- **DAFI Scholarship**, UNHCR — Awarded the DAFI scholarship to support undergraduate studies, 2019.

9. Publications

1. **Hosseini, G.**; Jalalian, K.; Torabi, A.; Azizi, M. (2026). Towards Efficient Abdominal Multi-Organ Segmentation: A Lightweight Hybrid UNet-SegFormer Network. Manuscript in preparation (First author).

2. Jalalian, K.; **Hosseini, G.**; Ghiasi, R.; Bosaghzadeh, A. (2025). Enhancing Breast Cancer Detection in Mammography with UNet++ - Deep Learning Approach. Frontiers in Biomedical Technologies, 12(4), 812–829**.** DOI: https://doi.org/10.18502/fbt.v12i4.19818

3. **Hosseini, G**.; Jalalian, K.; Ghiasi, R.; Bosaghzadeh, A. (2023). Breast Cancer Detection Using Deep Convolutional Neural Network: A Pre-processing-Driven Approach. 2023 30th National and 8th International Iranian Conference on Biomedical Engineering (ICBME), IEEE, Tehran, Iran. DOI: 10.1109/ICBME61513.2023.10488557 (First Author).

10. References

1) Alireza Bosaghzadeh, Computer Engineering
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2) Batol Bahraini, Electrical Engineering
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